

DAY 11

1. AI Recommendation System Research

AI Recommendations

AI Recommendation Topic	Explanation	Example for Fashion Platform
Personalized Fashion Feed	AI recommends items based on browsing history and interactions	Suggest outfits based on recently viewed products
Cross-Selling Recommendations	AI recommends complementary products	Suggest handbags with dresses
Color Preference Analysis	Tracks preferred colors and styles	Recommend pastel collections to users preferring light shades
Behavioral Recommendation Engine	Uses clicks, searches and purchases	Promote products matching shopping patterns
Seasonal Recommendation Logic	Shows products based on trends and seasons	Recommend festive collections during holidays
Cart Recovery Suggestions	Recommends products left in cart	Send reminders for abandoned items
Trend Prediction System	Predicts upcoming interests	Display trending outfits on homepage
Style Combination Suggestions	Suggests complete outfit sets	Match tops with accessories and footwear

Recommendation systems improve user experience by analyzing interactions, shopping history, category preference, and engagement behavior. AI helps increase personalization, product discovery, and conversion rates.

2. User Behavior Prediction Analysis

Prediction Analysis

Prediction Insight	Observation	Suggested Action
Highly active users show purchase intent	Long sessions and repeated visits indicate interest	Provide exclusive offers and recommendations
Users skipping wishlist feature may disengage	Low interaction users browse less	Promote wishlist reminders
Users viewing similar products repeatedly	Indicates comparison behavior	Show comparison recommendations
Short session users have higher churn risk	Lower engagement observed	Send re-engagement notifications
Users interact differently across categories	Preferences vary by style	Personalize homepage products

User behavior prediction helps identify user interests, engagement level, and future purchase possibility. Understanding user activity supports personalization and customer retention.

3. Virtual Try-On Analytics Task

Virtual Try-On Analytics

Virtual Try-On Topic	Observation	Importance
Why Users Use Try-On	Users check fit and appearance before purchase	Improves buying confidence
Most Used Features	Rotation view, outfit preview, styling options	Improves interaction
Purchase Influence	Users trust products more after trying virtually	Increases conversion
Return Reduction Impact	Users better understand product fit	Lowers return probability
User Engagement Impact	Interactive experience keeps users active	Improves retention

Benefits of Virtual Try-On

Benefit	Explanation
Better purchase decisions	Users evaluate products before buying
Increased interaction time	Users spend more time exploring products
Reduced uncertainty	Helps understand appearance and fit

Benefit	Explanation
Better product exploration	Users try multiple combinations
Improved purchase conversion	Confident customers purchase more

4. Product Performance Analysis

Product Performance Analysis

Insight	Observation	Suggested Action
Party collections receive maximum interactions	Users engage more with premium fashion items	Promote trending collections
Low-engagement products need visibility	Some products receive fewer clicks	Improve promotions and images
Traditional wear performs strongly	Users frequently view ethnic categories	Expand ethnic product range
Products with high try-on activity perform better	Users engaging with try-on purchase more	Encourage virtual try-on usage

Product analysis indicates that products with higher interaction and try-on activity generally achieve better conversion performance.

5. Smart Shopping Insights Dashboard

Smart Insights Dashboard

Metric	Insight
Top Performing Category	Party Wear
Highest Engagement Product	Premium Dress Collection
Average User Interaction	High
Users with High Purchase Probability	4 Users
Major Retention Concern	Low activity users

Dashboard insights help identify customer preference patterns, engagement trends, and purchase opportunities.

6. Data Visualization Task

Chart 1 — Product Preference Graph

Most Preferred Products Analysis

The chart compares product popularity based on views and engagement levels. Premium and seasonal collections received the highest user interest while workwear products showed comparatively lower interactions.

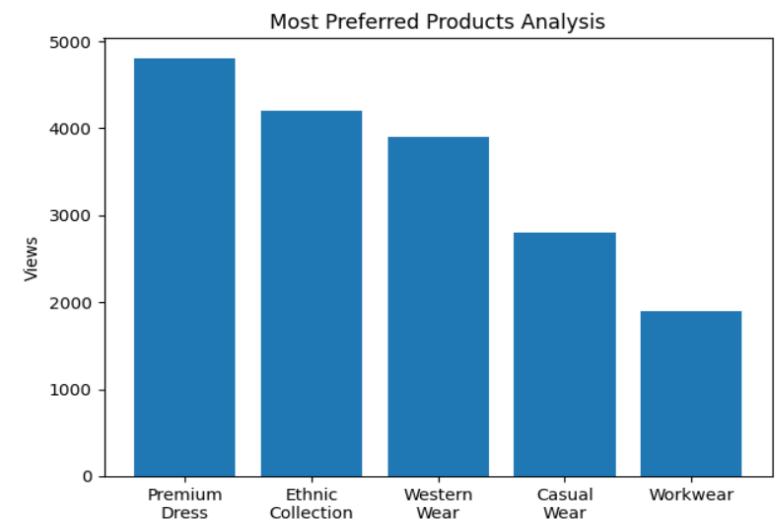


Chart 2 — Engagement vs Purchase Graph

Interaction vs Conversion Analysis

The chart compares user interaction with purchase behavior. Products with higher engagement and try-on usage generally show stronger purchase performance.

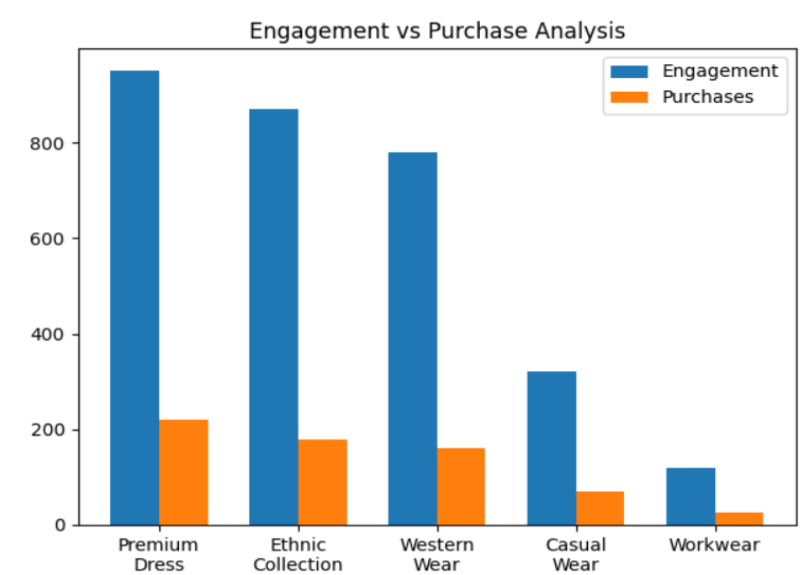
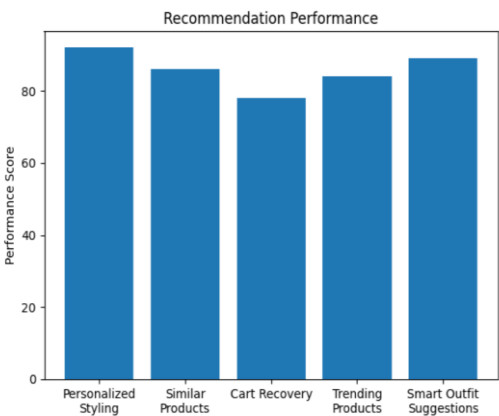


Chart 3 — Recommendation Analytics Graph

Recommendation Type	Performance Score
Personalized Styling	92
Similar Product Suggestions	86
Cart Recovery Recommendations	78
Trending Products	84
Smart Outfit Suggestions	89



Recommendation analytics indicate personalized suggestions achieve the highest engagement performance.

Recommendation Performance

The recommendation performance graph shows personalized recommendations and outfit suggestions generating higher user interaction compared to other recommendation methods. AI-based personalization significantly improves engagement and shopping experience.

Conclusion

This task focused on analyzing AI recommendation systems, user behavior prediction, virtual try-on analytics, product performance, and smart shopping insights for fashion e-commerce platforms.

The study showed that AI-driven recommendations improve personalization by suggesting relevant products based on user interests and behavior. User behavior analysis helped identify engagement patterns, purchase intent, and retention risks, supporting better decision-making.

Virtual try-on analytics highlighted its role in improving customer confidence, increasing engagement, reducing product returns, and enhancing the shopping experience. Product performance analysis also showed that highly engaged products generally achieve better purchase results.

Overall, this task helped in understanding how AI recommendations, user analytics, virtual try-on technology, and smart dashboards can improve customer experience, engagement, retention, and conversion performance in fashion e-commerce platforms.